HARSHA S. BHAT

CNRS Director of Research & Teaching Professor

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EDUCATION

École Normale Supérieure, France	H. D. R.†	Supershear Earthquakes	2021/01
Harvard University, USA	Ph. D.*	Mechanical Sciences	2007/06
Harvard University, USA	M. S.	Engineering Sciences	2002/06
NITK, India	B. E.	Civil Engineering	2001/06

† *Habilitation à Diriger des Recherches* * *Supervised by Prof. J. R. Rice & Dr. R. Dmowska*

CURRENT POSITION

École Normale Supérieure, France	2024/10 ▶ Present	CNRS Director of Research
Ecole Polytechnique, France	2022/09 ▶ Present	Teaching Professor (PCC)

PAST POSITIONS

École Normale Supérieure, France	2016/05 ▶ 2024/09	CNRS Research Scientist
NISER, India	2021/11 ▶ 2023/11	Visiting Professor
Institut de Physique du Globe de Paris, France	2012/01 ▶ 2016/05	CNRS Research Scientist
University of Southern California, USA	2010/03 ▶ 2011/12	Asst. Professor (Research)
University of Southern California, USA	2007/11 ▶ 2010/03	Post Doctoral Fellow
California Institute of Technology, USA	2007/11 ▶ 2010/03	Visitor in Aeronautics
Harvard University, USA	2007/05 ▶ 2007/10	Post Doctoral Fellow
Harvard University, USA	2001/11 ▶ 2007/05	Grad. Research Associate

FUNDING & GRANTS†

- 2021–2025 ▶ 2M€ ERC Consolidator Grant, PERSISMO (Grant No. 865411)
- 2018–2018 ▶ 25k€ ENS Actions Incitatives
- 2017–2017 ▶ 6k€ TelluS INSU - action ALEAS
- 2023–2027 ▶ 100k€ ANR SMEC
- 2023–2026 ▶ 100k€ ANR Univ. Tokyo SESAME
- 2023–2025 ▶ 66k€ SPARC Grant India

HONORS AND AWARDS

- 2018 CNRS Award for Doctoral Supervision and Research
- 2018 Grand Prix Michel Gouilloud Schlumberger, French Academy of Sciences
- 2006 Harvard University Certificate of Distinction in Teaching
- 2004 Harvard University Certificate of Distinction in Teaching
- 2003 Harvard University Certificate of Distinction in Teaching

CURRENT TEACHING RESPONSIBILITIES AT ECOLE POLYTECHNIQUE†

Mécanique du Milieu continus Mécanique des Solides Mécanique de la Rupture Mécanique des matériaux et des structures

STUDENTS & POSTDOCS

Postdoctoral Associates

Ankit Gupta (India)	2024-2026	
Suli Yao (China)	2025-2027	
Navid Kheirdast (Iran)	2022-2025	Post Doctoral fellow at ISTEP
Michelle Almakari (France)	2021–2023	
Carlos D. Villafuerte (Mexico)	2021–2023	Asst. Prof. UNAM
Ekeabino Momoh (Nigeria)	2019–2022	AXA Postdoc Fellow
Lucile Bruhat (France)	2018–2021	Natural Catastrophe Risk Analyst at AXA
Lisa Gordeliy (Russia)	2019–2019	Senior Engineer ResFrac
Marion Y. Thomas (France)	2014–2016	CNRS Scientist at Université de Rennes

PhD Students

Bharath Shanmugasundaram (India)	2024–2027	
Yishuo Zhou (China)	2024–2027	
Thomas Melkior (France)	2023–2026	
Caiyuan Fan (China)	2023–2027	
Jinhui Cheng (China)	2021–2024	Postdoc Caltech
Augustin Thomas (France)	2020–2023	Research Scientist, BRGM
Joseph Flores Cuba (Peru)	2020–2023	Data Scientist PowerBI
Claudia Hulbert (France)	2018–2021	CEO Geolabe
Samson Marty (France)	2017–2020	Postdoc at Caltech
Marshall A. Martinez (USA)	2014–2019	Engineer at Joby Aviation
Kurama Okubo (Japan)	2015–2018	Research Scientist at NIED, Japan
Pierre Romanet (France)	2014–2017	Postdoc GeoAzur, Nice
Vahe Gabuchian (USA)	2010–2015	Research Scientist at Caltech
François X. Passelègue (France)	2011–2014	CNRS Scientist at GeoAzur, Nice
Jonathan Mihaly (USA)	2008–2013	Jet Propulsion Laboratory
Michael Mello (USA)	2007–2012	Teaching Professor at Caltech

Undergraduate and Masters Interns

Valentin Marnat	2022	Phillipe Danre	2017
Roxane Ferry	2021	Eleni Kolokytha	2015
Jinhui Cheng	2020	Victor Barolle	2015
Phillipe Danre	2019	Kurama Okubo	2014
Roxane Ferry	2019	Thibaut Perol	2013
Hugo Lestrelin	2019	Lucile Bruhat	2012
Nicolas Mercury	2018	Marion Olives	2004
Luc Illien	2018	Sonia Fliss	2003

PAST TEACHING ACTIVITIES†

1) Mécanique des Milieux Continus
2) Active Faults : Geometry
3) Seismic Ruptures and Scaling Laws
4) Introduction to Rock Physics
5) Mathematical Methods in the Sciences
6) Environmental Risks and Disasters
7) Ordinary and Partial Differential Equations
8) Complex and Fourier Analysis
9) Computational Solid and Structural Mechanics
10) Solid Mechanics
11) Introduction to the Mechanics of Solids
12) Mechanics of Fracture
13) Advanced Geomechanics
14) Mécanique de la Fracturation

† *Classes taught with various colleagues at Harvard, Caltech, IPGP, ENS*

ORGANIZATION OF SCIENTIFIC MEETINGS

- Apr 2024: Across the time scales, from earthquakes to earthquake cycle: EGU 2024
- Apr 2023: Across the time scales, from earthquakes to earthquake cycle: EGU 2023
- Jun 2019: Coupled Processes In Fracture Propagation In Geo-Materials: From Hydraulic Fractures To Earthquakes: CISM Advanced School, Udine, Italy
- Apr 2015: Seismological Society of America, Multiscale Modeling and Characterization of Fragmentation and Damage Patterns in Fault Zones
- Dec 2014: American Geophysical Union, Fault Zone Properties And Processes During Dynamic Ruptures

INSTITUTIONAL RESPONSIBILITIES

- 2018–2024: Team Leader of Faults & Earthquakes Group, ENS
- 2018–2019: Co-organizer of the Internal Seminar, ENS

REVIEWING ACTIVITIES

American Geophysical Union Seismological Society of America International Journal of Fracture Geological Society of America Science Nature Journal of the Mechanics and Physics of Solids European Journal of Mechanics - A/Solids Earth and Planetary Science Letters Geophysical Research Letters Journal of Structural Geology Proceedings of the National Academies of Science, USA Geology Geophysical Journal International Journal of Applied Mechanics National Science Foundation European Research Council Nature Communications Nature Geoscience Science Advances

BOOKS

Thomas, M. Y., Mitchell, T. M., **Bhat, H. S.,** eds. (2017b). "*Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227*". American Geophysical Union (AGU). DOI: 10.1002/9781119156895.

Bizzarri, A., **Bhat, H. S.,** eds. (2012). "*The mechanics of faulting: From laboratory to earthquakes*". Research Signpost.

BOOK CHAPTERS

Thomas, M. Y., **Bhat, H. S.,** (2022b). "*Loi de friction et modélisation numérique du cycle sismique*". in Le Cycle Sismique. Ed. by F. Rolandone. ISTE-Wiley.

Thomas, M. Y., **Bhat, H. S.,** (2022a). "*Friction Laws and Numerical Modeling of the Seismic Cycle*". in The Seismic Cycle: From Observation to Modeling. Ed. by F. Rolandone. ISTE-Wiley.

MANUSCRIPTS

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Under Review or Accepted

Almakari, M., Kheirdast, N., Villafuerte, C. D., Thomas, M. Y., Dubernet, P., Cheng, J., Gupta, A., Romanet, P., Chaillat, S., **Bhat, H. S.,** (2025). "*Spectrum of slip dynamics, scaling & statistical laws emerge from simplified model of fault and damage zone architecture*". subm. J. Geophys. Res. arXiv: 2509.04909.

Melkior, T., **Bhat, H. S.**, Amlani, F., (2025). "*Tsunami modeling with dynamic seafloors: a high-order solver validated with shallow water benchmarks*". J. Comput. Phys. submitted. arXiv: 2508.20596.

Michel, S., Scotti, O., Hok, S., **Bhat, H. S.**, Kheirdast, N., Romanet, P., Almakari, M., Cheng, J., (2025). "*Probability of earthquake fault jumps from physics based criterion*". J. Geophys. Res. arXiv: 2501.15948.

Published

Latour, S., Lebihain, M., **Bhat, H. S.**, Twardzik, C., Bletery, Q., Hudnut, K. W., Passelègue, F., (2025). "*Direct Estimation of Earthquake Source Properties from a Single CCTV Camera*". Science 390, pp. 463–467. DOI: 10.1126/science.adz1705. arXiv: 2505.15461v2.

Cheng, J., Almakari, M., Peruzzo, C., Lecampion, B., **Bhat, H. S.,** (2025). "*FASTDASH, a Quasi-dynamic 3D Seismic Cycle Model by Using Boundary Element Method with H-matrices*". Geophys. J. Int. DOI: 10.1093/gji/ggaf230. arXiv: 2505.02398v3.

Momoh, E., **Bhat, H. S.**, Tait, S., Gerbault, M., (2025). "*Volumetric (dilatant) plasticity in geodynamic models and implications on thermal dissipation and strain localization*". Geophys. J. Int. 240.3, pp. 1551–1578. DOI: 10.1093/gji/ggae463.

Ferry, R., Thomas, M. Y., **Bhat, H. S.**, Dubernet, P., (2025). "*Depth Dependence of Coseismic Off-Fault Damage and its Effects on Rupture Dynamics*". J. Geophys. Res. e2024JB029787. DOI: 10.1029/2024JB029787.

Petit, L., Olive, J.-A., Schubnel, A., Le Pourhiet, L., **Bhat, H. S.,** (2024). "*A brittle constitutive law for long-term tectonic modeling based on sub-critical crack growth*". to appear in Geochem. Geophys. Geosyst. 25, e2023GC011229. DOI: 10.1029/2023GC011229.

Jeandet-Ribes, L., Thomas, M. Y., **Bhat, H. S.,** (2023). "*On the importance of setting 3-D stress field in simulations of on- and off-fault deformation*". Geophys. J. Int. 235.3, pp. 2962–2978. DOI: 10.1093/gji/ggad401.

Marty, S., Schubnel, A., **Bhat, H. S.**, Aubry, J., Fukuyama, E., Latour, S., Nielsen, S., Madariaga, R., (2023). "*Nucleation of laboratory earthquakes: quantitative analysis and scalings*". J. Geophys. Res. 128.e2022JB026294. DOI: 10.1029/2022JB026294.

Amlani, F., **Bhat, H. S.**, Simons, W. J. F., Schubnel, A., Vigny, C., Rosakis, A. J., Efendi, J., Elbanna, A., Dubernet, P., Abidin, H. Z., (2022). "*Supershear shock front contribution to the tsunami from the 2018 Mw 7.5 Palu, Indonesia earthquake*". Geophys. J. Int. 230, pp. 2089–2097. DOI: 10.1093/gji/ggac162.

Jara, J., Bruhat, L., Thomas, M. Y., Antoine, S., Okubo, K., Klinger, Y., Jolivet, R., **Bhat, H. S.,** (2021). "*Signature of transition to supershear rupture speed in coseismic off-fault damage zone*". Proc. R. Soc. A. 477, p. 20210364. DOI: 10.1098/rspa.2021.0364.

Elbanna, A., Abdelmeguid, M., Ma, X., Amlani, F., **Bhat, H. S.**, Synolakis, C., Rosakis, A. J., (2021). "*Anatomy of Strike Slip Fault Tsunami Genesis*". Proc. Natl. Acad. Sci. USA. DOI: 10.1073/pnas.2025632118.

Bhat, H. S. (2021). "*Supershear Earthquakes*". PhD thesis. Habilitation à Diriger des Recherches, Ecole Normale Supérieure.

Jeandet-Ribes, L., Cubas, N., **Bhat, H. S.**, Steer, P., (2020). "*Response of a single fault to transient normal stress change, and implications of large erosional events on the seismic cycle*". Geophys. Res. Lett. 47.e2020GL087631. DOI: 10.1029/2020GL087631.

Jolivet, R, Simons, M, Duputel, Z, Olive, J.-A., **Bhat, H. S.**, Bletery, Q., (2020). "*Interseismic Loading of Subduction Megathrust Drives Long-Term Uplift in Northern Chile*". Geophys. Res. Lett. 47.8, e2019GL085377. DOI: 10.1029/2019GL085377.

Okubo, K., Rougier, E, Lei, Z., **Bhat, H. S.,** (2020). "*Modeling earthquakes with off-fault damage using the combined finite discrete element method*". J. Comp. Part. Mech. DOI: 10.1007/s40571-020-00335-4.

Okubo, K., **Bhat, H. S.**, Rougier, E., Marty, S., Schubnel, A., Lei, Z., Knight, E. E., Klinger, Y., (2019). "*Dynamics, radiation and overall energy budget of earthquake rupture with coseismic off-fault damage*". J. Geophys. Res. 124. DOI: 10.1029/2019JB017304.

Marty, S., Passelègue, F. X., Aubry, J., Schubnel, A., **Bhat, H. S.**, Madariaga, R., (2019). "*Origin of high-frequency radiation during laboratory earthquakes*". Geophys. Res. Lett. 46. DOI: 10.1029/2018GL080519.

Aubry, J., Passelègue, F. X., Deldicque, D., Girault, F., Marty, S., Lahfid, A., **Bhat, H. S.**, Escartin, J., Schubnel, A., (2018). "*Frictional heating processes and energy budget during laboratory earthquakes*". Geophys. Res. Lett. 45. DOI: 10.1029/2018GL079263.

Klinger, Y. (2018). "*Earthquake damage patterns resolve complex rupture processes*". Geophys. Res. Lett. DOI: 10.1029/2018GL078842.

Cruz-Atienza, V. M., Villafuerte, C. D., **Bhat, H. S.,** (2018). "*Rapid tremor migration and pore-pressure waves in subduction zones*". Nat. Commun. 9.1, p. 2900. DOI: 10.1038/s41467-018-05150-3.

Thomas, M. Y., **Bhat, H. S.,** (2018). "*Dynamic evolution of off-fault medium during an earthquake: a micromechanics based model*". Geophys. J. Int. 214.2, pp. 1267–1280. DOI: 10.1093/gji/ggy129.

Romanet, P., **Bhat, H. S.**, Jolivet, R., Madariaga, R., (2018). "*Fast and slow earthquakes emerge due to fault geometrical complexity*". Geophys. Res. Lett. DOI: 10.1029/2018GL077579.

Gabuchian, V., Rosakis, A. J, **Bhat, H. S.**, Madariaga, R., Kanamori, H., (2017). "*Experimental evidence that thrust earthquake ruptures might open faults*". Nature 545.336–339. DOI: 10.1038/nature22045.

Thomas, M. Y., **Bhat, H. S.**, Klinger, Y., (2017a). "*Effect of Brittle off-fault Damage on Earthquake Rupture Dynamics*". in Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227. Ed. by M. Y. Thomas, H. S. Bhat, and T. M. Mitchell. American Geophysical Union (AGU), pp. 255–280. DOI: 10.1002/9781119156895.ch14.

Passelègue, F. X., Latour, S., Schubnel, A., Nielsen, S., **Bhat, H. S.**, Madariaga, R., (2017). "*Precursory Processes during Laboratory Earthquakes*". in Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture, Geophysical Monograph 227. Ed. by M. Y. Thomas, H. S. Bhat, and T. M. Mitchell. American Geophysical Union (AGU). Chap. 12, pp. 229–242. DOI: 10.1002/9781119156895.ch12.

Perol, T., **Bhat, H. S.,** (2016). "*Micromechanics based permeability evolution in brittle materials at high strain rates*". Pure Appl. Geophys. Pp. 1–12. DOI: 10.1007/s00024-016-1354-4.

Passelègue, F. X., Schubnel, A., Nielsen, S., **Bhat, H. S.**, Deldicque, D., Madariaga, R., (2016). "*Dynamic rupture processes inferred from laboratory microearthquakes*". J. Geophys. Res. 121. DOI: 10.1002/2015JB012694.

Mello, M., **Bhat, H. S.**, Rosakis, A. J., (2016). "*Spatiotemporal properties of sub-Rayleigh and supershear rupture velocity fields : Theory and Experiments*". J. Mech. Phys. Solids 93, pp. 153–181. DOI: 10.1016/j.jmps.2016.02.031.

Vallage, A, Klinger, Y, Grandin, R, **Bhat, H. S.**, Pierrot-Deseilligny, M, (2015). "*Inelastic surface deformation during the 2013 Mw 7.7 Balochistan, Pakistan, earthquake*". Geology 43.12, pp. 1079–1082. DOI: 10.1130/G37290.1.

Frank, W. B., Shapiro, N. M., Husker, A. L., Kostoglodov, V, **Bhat, H. S.**, Campillo, M, (2015). "*Along-fault pore-pressure evolution during a slow-slip event in Guerrero, Mexico*". Earth Planet. Sc. Lett. 413, pp. 135–143. DOI: 10.1016/j.epsl.2014.12.051.

Siriki, H., **Bhat, H. S.**, Lu, X., Krishnan, S., (2015). "*A Laboratory Earthquake-Based Stochastic Seismic Source Generation Algorithm for Strike-Slip Faults*". Bull. Seism. Soc. Am. 105.4, pp. 2250–2273. DOI: 10.1785/0120140110.

Mello, M., **Bhat, H. S.**, Rosakis, A. J., Kanamori, H., (2014). "*Reproducing The Supershear Portion Of The 2002 Denali Earthquake Rupture In Laboratory*". Earth Planet. Sc. Lett. 387, pp. 89–96. DOI: 10.1016/j.epsl.2013.11.030.

Passelègue, F. X., Schubnel, A., Nielsen, S., **Bhat, H. S.**, Madariaga, R., (2013). "*From Sub-Rayleigh to Supershear Ruptures During Stick-Slip Experiments on Crustal Rocks*". Science 340.6137, pp. 1208–1211. DOI: 10.1126/science.1235637.

Bhat, H. S., Rosakis, A. J., Sammis, C. G., (2012). "*A Micromechanics Based Constitutive Model For Brittle Failure at High Strain Rates*". J. Appl. Mech. 79.3. DOI: 10.1115/1.4005897.

Bhat, H. S., Sammis, C. G., Rosakis, A. J., (2011). "*The Micromechanics of Westerley Granite at Large Compressive Loads*". Pure Appl. Geophys. 168.12, pp. 1–18. DOI: 10.1007/s00024-011-0271-9.

Bhat, H. S., Biegel, R. L., Rosakis, A. J., Sammis, C. G, (2010). "*The Effect of Asymmetric Damage on Dynamic Shear Rupture Propagation II: With Mismatch in Bulk Elasticity*". Tectonophysics 493.3, pp. 263–271. DOI: 10.1016/j.tecto.2010.03.016.

Biegel, R. L., **Bhat, H. S.**, Sammis, C. G., Rosakis, A. J., (2010). "*The Effect of Asymmetric Damage on Dynamic Shear Rupture Propagation I: No Mismatch in Bulk Elasticity*". Tectonophysics 493.3, pp. 254–262. DOI: 10.1016/j.tecto.2010.03.020.

Mello, M., **Bhat, H. S.**, Rosakis, A. J., Kanamori, H., (2010). "*Identifying the unique ground motion signatures of supershear earthquakes: Theory and experiments*". Tectonophysics 493, pp. 297–326. DOI: 10.1016/j.tecto.2010.07.003.

Templeton, E. L., **Bhat, H. S.**, Dmowska, R., Rice, J. R., (2010). "*Dynamic rupture through a branched fault configuration at Yucca Mountain and resulting ground motions*". Bull. Seism. Soc. Am. 100.4, pp. 1485–1497. DOI: 10.1785/012009012110.1785/0120090121.

Harris, R. A. (2009). "*The SCEC/USGS dynamic earthquake rupture code verification exercise*". Seismol. Res. Lett. 80.1. DOI: 10.1785/gssrl.80.1.119.

Sammis, C. G., Rosakis, A. J., **Bhat, H. S.,** (2009). "*Effects of Off-fault Damage on Earthquake Rupture Propagation: Experimental Studies*". Pure Appl. Geophys. 166. DOI: 10.1007/s00024-009-0512-3.

Templeton, E. L., Baudet, A., **Bhat, H. S.**, Dmowska, R., Rice, J. R., Rosakis, A. J., Rousseau, C. E., (2009). "*Finite Element Simulations of Dynamic Shear Rupture Experiments and Dynamic Path Selection Along Kinked and Branched Faults*". J. Geophys. Res. B08304. DOI: 10.1029/2008JB006174.

Dunham, E. M., **Bhat, H. S.,** (2008). "*Attenuation of radiated ground motion and stresses from three-dimensional supershear ruptures*". J. Geophys. Res. 113.B08319. DOI: 10.1029/2007JB005182.

Bhat, H. S., Dmowska, R., King, G. C. P., Klinger, Y., Rice, J. R., (2007a). "*Off-fault damage patterns due to supershear ruptures with application to the 2001 M_w 8.1 Kokoxili (Kunlun) Tibet earthquake*". J. Geophys. Res. B06301. DOI: 10.1029/2006JB004425.

Bhat, H. S., Olives, M., Dmowska, R., Rice, J. R., (2007b). "*Role of fault branches in earthquake rupture dynamics*". J. Geophys. Res. B11309. DOI: 10.1029/2007JB005027.

Bhat, H. S. (2007). "*Role of Geometric Complexities and Off-Fault Damage in Dynamic Rupture Propagation*". PhD thesis. Harvard University.

Fliss, S., **Bhat, H. S.**, Dmowska, R., Rice, J. R., (2005). "*Fault branching and rupture directivity*". J. Geophys. Res. B06312. DOI: 10.1029/2004JB003368.

Bhat, H. S., Dmowska, R., Rice, J. R., Kame, N., (2004). "*Dynamic slip transfer from the Denali to Totschunda faults, Alaska: Testing theory for fault branching*". Bull. Seism. Soc. Am. 94, S202–S213. DOI: 10.1785/0120040601.